

MAE 5500 Aerodynamics : Wing Warm-Up Code

W1. Write a Python 3 computer program that uses the series solution to Prandtl's lifting-line equation to numerically predict the lift and induced-drag coefficients for unswept elliptic and tapered wings having no geometric or aerodynamic twist.

1. The program must give the user the opportunity to specify the number of nodes per semispan. One node must be placed at the midspan, one node must be placed at the wingtip, and the nodes must be spaced using cosine clustering. The node at the midspan is counted on both semispans so that the total number of nodes is always odd. For example, if the user specifies 50 nodes per semispan, a total of 99 terms will be carried in the Fourier sine series.
2. The program must allow the user to specify the wing geometry (aspect ratio and taper ratio if the wing is tapered, and aspect ratio if the wing is elliptic). The user must also be able to specify the airfoil section lift slope. This information must be read in from a JSON file (with file name "W1_input.json") of the format

```
{
  "wing" : {
    "planform" : {
      "type" : "tapered",
      "type_options" : [["elliptic: (requires aspect_ratio)"],
                        ["tapered: (requires aspect_ratio and taper_ratio)"]],
      "aspect_ratio" : 8.0,
      "taper_ratio" : 0.5
    },
    "airfoil_lift_slope" : 6.283185307179590,
    "nodes_per_semispan" : 4
  },
  "condition" : {
    "alpha_root[deg]" : 5.0
  },
  "view" : {
    "planform" : true
  }
}
```

3. The program must write the $[C]$ and $[C]^{-1}$ matrices as well as the Fourier coefficients $\{a_n\}$ to a formatted file named "Solution.txt" with one complete row per line. Sample solutions for these matrices and arrays are available [here](#) for the input values shown above.
4. The program must write κ_L , κ_D , span efficiency factor e_s , and the wing lift slope $C_{L,\alpha}$ to the screen, as well as the resulting lift and induced-drag coefficients for the operating angle of attack.
5. The program must produce a plot of the wing planform. The geometry must be normalized by wingspan and show the location of the lifting line, leading edge, trailing edge, and locations where the series solution is satisfied. An example plot is included [here](#).
6. Use your program to rework Exercise Set problems 6.1, 6.2, and 6.3 with $nodes_per_semispan = 50$ and make sure your code matches your previous results based on the plots.
7. Your code should be entirely contained in a single file called "LastName_FirstName.py" that can be run by typing "python" and the filename from a command line. Submit this file online to hand in your homework.

W2. Make a copy of your code for problem W1 and modify the code to use lifting-line theory to predict the lift and induced drag for elliptic and tapered wings with twist.

1. The input to the code should be a JSON file (with file name "W2_input.json") of the format

```
{
  "wing" : {
    "planform" : {
      "type" : "tapered",
      "type_options" : [["elliptic: (requires aspect_ratio)"],
                        ["tapered: (requires aspect_ratio and taper_ratio)"]],
      "aspect_ratio" : 8.0,
      "taper_ratio" : 0.5
    },
    "airfoil_lift_slope" : 6.283185307179590,
    "nodes_per_semispan" : 4,
    "washout" : {
      "distribution" : "optimum",
      "distribution_options" : [["none: no twist"],
                                ["linear: linear twist distribution"],
                                ["optimum: optimum twist distribution"]],
      "amount[deg]" : 5.0,
      "amount_options" : [["value: real number in degrees"],
                           ["optimum: requires CL_design"]],
      "CL_design" : 0.8
    }
  },
  "condition" : {
    "alpha_root[deg]" : 5.0
  },
  "view" : {
    "planform" : true,
    "washout_distribution" : true
  }
}
```

2. The code must compute the washout distribution and amount based on the inputs from the file. The solution for the $\{b_n\}$ array must also be written to the "Solution.txt" file.

3. The program must also write ϵ_Ω , κ_{DL} , and $\kappa_{D\Omega}$ to the screen.

4. The program must also produce a plot of the washout distribution $\omega(z/b)$.

5. Test your program for linear twist using the inputs given above and compare to the solution provided [here](#).

6. Test your program for optimum twist using the following changes to the input file

```
{
  "wing" : {
    "washout" : {
      "distribution" : "optimum"
    }
  }
}
```

and compare your results to those provided [here](#).

7. Use your code to plot the induced-drag coefficient as a function of lift coefficient for an elliptic wing with an aspect ratio of 8 with no washout using 50 nodes per semispan. Repeat this process for the same wing with 5 degrees linear washout. Compare the results of the two wings by overlaying the solutions on a single plot and comment on the differences.

8. Use your code to plot the induced-drag coefficient as a function of lift coefficient for a tapered wing with an aspect ratio of 8, taper ratio of 0.5, and no washout using 50 nodes per semispan. Repeat this process for the same wing with 5 degrees linear washout. Compare the results of the two wings by overlaying the solutions on a single plot and comment on the differences.

9. Submit results for 7 and 8 as a PDF document online to hand in your homework.

W3. Make a copy of your code for problem W2 and modify the code to include the effects of aileron deflection and roll.

1. The input to the code should be a JSON file (with file name "W3_input.json") of the format

```
{
  "wing" : {
    "planform" : {
      "type" : "elliptic",
      "planform_type_notes" : [["elliptic: (requires aspect_ratio)"],
                              ["tapered: (requires aspect_ratio and taper_ratio)"]],
      "aspect_ratio" : 8.0,
      "taper_ratio" : 0.5
    },
    "airfoil_lift_slope" : 6.283185307179590,
    "nodes_per_semispan" : 4,
    "washout" : {
      "distribution" : "linear",
      "distribution_options" : [["none: no twist"],
                               ["linear: linear twist distribution"],
                               ["optimum: optimum twist distribution"]],
      "amount[deg]" : 5.0,
      "amount_options" : [["value: real number in degrees"],
                          ["optimum: requires CL_design"]]
    },
    "aileron" : {
      "begin[z/b]" : 0.22,
      "end[z/b]" : 0.45,
      "begin[cf/c]" : 0.18,
      "end[cf/c]" : 0.18,
      "hinge_efficiency" : 0.85
    }
  },
  "condition" : {
    "alpha_root[deg]" : 5.0,
    "aileron_deflection[deg]" : 5.0,
    "pbar" : "steady",
    "pbar_notes" : "value or 'steady'"
  },
  "view" : {
    "planform" : true,
    "washout_distribution" : true,
    "aileron_distribution" : true
  }
}
```

2. The code must compute an aileron distribution such that the hinge line is straight from the root to the tip of the aileron.

3. The flap-chord fraction c_f/c must be computed at each spanwise location of interest. This result must be used to compute the section flap effectiveness assuming a deflection efficiency of 1.0.

4. The solution for the $\{c_n\}$ and $\{d_n\}$ arrays must also be written to the "Solution.txt" file.

5. The program must also write C_{ℓ, δ_a} and $C_{\ell, \dot{p}}$ to the screen, along with the rolling-moment coefficient C_ℓ and yawing-moment coefficient C_n for the operating condition.
6. The program must compute the induced drag two ways. The first from Eq. (1.8.25) which neglects aileron and rolling rate. The second from Eq. 6.6 in the Handbook, which includes aileron deflection and rolling rate.
7. The program must display the aileron geometry on the "planform" plot, and produce a new plot showing the aileron distribution $\chi(z/b)$.
8. Test your program using the inputs given above and compare to the solution provided [here](#).
9. Use your code to compute the lift coefficient and induced-drag coefficient on a tapered wing with a taper ratio of 0.5, aspect ratio of 8, using 50 nodes per semispan and otherwise identical properties to those given in this example. Submit your results including plots and solution for your homework assignment.